

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA51

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO5: *Develop the network security strategies based on the study of viruses, worms, firewall architectures, and IDS/IPS functionalities.* CO (BL5, BL6)

Assessment Tool Weightage: 10%

QUESTIONS: 20 TOTAL MARKS: 20 Annexure A MOCK INTERVIEW

Code of Conduct:

I P.GOWTHAM REDD certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.Gowtham Reddy

Annexure A

MOCK INTERVIEW

1. Which would you prioritize in a secure network design: **performance or security**, and how would you balance both?

Security should be prioritized, but performance must also be maintained. Use optimized security controls, load balancing, and efficient monitoring to achieve both.

2. Can you explain the difference between **stateful and stateless firewalls**, and which one is more suitable for enterprise networks?

A **stateful firewall** tracks active connections, while a stateless firewall checks each packet independently. Stateful firewalls are generally more suitable for enterprise networks.

3.If an IDS generates too many false positives, how would you optimize its performance?

Tune detection rules, update signatures, establish accurate thresholds, and whitelist trusted activities. Regularly analyze alerts to improve detection accuracy.

4.How would you design a **real-time intrusion detection system** for high-speed networks?

Use high-performance packet processing, parallel analysis, signature/anomaly detection, and distributed sensors. Alerts should be generated with minimal delay.

5.What are the key challenges in detecting **zero-day malware**, and how would you address them?

Zero-day malware has no known signature, making traditional detection difficult. Behavioral analysis, sandboxing, machine learning, and threat intelligence can help detect it.

6. How would you differentiate between **signature-based and anomaly-based IDS**, and when would you use each?

Signature-based IDS detects known attacks using predefined patterns. **Anomaly-based IDS** identifies deviations from normal behavior and is useful for detecting unknown attacks.

7.If a network is under a **DDoS attack**, what immediate and long-term mitigation strategies would you apply?

Immediately filter malicious traffic, rate-limit requests, and use DDoS protection services. Long-term measures include traffic monitoring, redundancy, CDN/Anycast, and scalable infrastructure.

8..How would you secure **remote access** to an enterprise network?

Use VPN or Zero Trust Network Access with strong authentication such as MFA. Apply encryption, least-privilege access, and continuous monitoring.

9.What are the limitations of **traditional antivirus software**, and how can modern solutions overcome them?

Traditional antivirus mainly relies on known signatures and may miss new or polymorphic malware. Modern solutions use behavior analysis, AI/ML, sandboxing, and endpoint detection.

10.How would you design a **secure email filtering system** to prevent phishing and spam attacks?

Use spam filters, malware scanning, URL analysis, attachment sandboxing, and phishing detection. SPF, DKIM, and DMARC should also be implemented.

11.If an attacker bypasses firewall rules, what additional layers of security would you implement?

Add multiple security layers such as IDS/IPS, endpoint protection, network segmentation, MFA, encryption, and SIEM-based monitoring using a **defense-in-depth** approach.

12.How would you evaluate the effectiveness of **network segmentation** in preventing cyberattacks?

Check whether segmentation limits attacker movement between systems. Measure blocked unauthorized connections, reduced lateral movement, and improved incident containment.

13.What role does **encryption play in preventing malware attacks**, if any?

Encryption does not directly prevent malware. However, it protects sensitive data from being stolen or intercepted, reducing the impact of malware attacks.

14.How would you secure a **cloud-based network infrastructure**?

Use IAM with least privilege, MFA, encryption, security groups, firewalls, continuous monitoring, vulnerability scanning, and secure configurations.

15.What is the importance of **log analysis and monitoring** in network security?

Logs provide evidence of suspicious activities and security incidents. Continuous monitoring helps detect attacks early and supports investigation and incident response.

16.How would you respond to a **security breach in a corporate network**?

Identify and contain the attack, isolate affected systems, remove the threat, recover from clean backups, and monitor for recurrence. Finally, perform a post-incident review.

17.What are the advantages and limitations of **next-generation firewalls (NGFW)**?

NGFWs provide application awareness, intrusion prevention, and advanced traffic inspection. They are more complex, expensive, and may reduce performance due to deep inspection.

18.How would you integrate **IDS/IPS with SIEM systems** for better threat detection?

Send IDS/IPS alerts and network logs to the SIEM. The SIEM correlates events from multiple sources, identifies attack patterns, and generates centralized alerts.

19.What are the best practices for **patch management and vulnerability assessment**?

Regularly scan systems for vulnerabilities, prioritize critical risks, test patches, and deploy them promptly. Maintain an updated inventory and verify successful patching.

20.How would you design a **future-proof network security architecture** considering emerging threats?

Use **Zero Trust, defense-in-depth, AI-based detection, cloud security, strong encryption, continuous monitoring, and automated response**. Regularly update the architecture according to emerging threats.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in	Applies concepts with moderate accuracy	Limited problemsolving ability; requires guidance	Unable to solve or apply concepts

		real scenarios			
Analytical Thinking	15	Analyzes questions critically and provides wellstructured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively

Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
------------	----	---	---	---------------------------------------	--------------------------------------